Thermodynamics

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1 Energy and Enthalpy

1.1 Potential and Kinetic Energy

Thermodynamics focuses on the **energy changes** that occur during chemical reactions and phase transitions. It allows us to predict whether a reaction will occur spontaneously and how much energy will release or absorb. In previous sections, we have touched briefly on certain thermodynamic concepts and terms, such as **endothermic and exothermic reactions**. In this section, we will dive deeper into these same concepts and term.

Energy in the context of chemistry can be boiled down to two main categories: **kinetic energy** and **potential energy**.

KINETIC ENERGY simply pertains to the motion of a molecule.

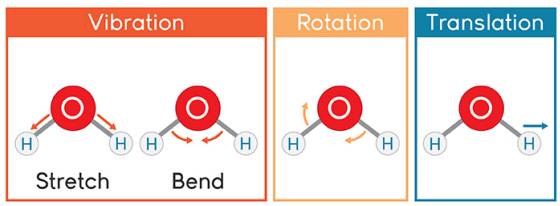

Figure 1: Kinetic Energy

POTENTIAL ENERGY (PE) pertains to the energy stored within the bond of intra- and intermolecular interactions. The basis of these interactions is electrostatic attractions that occur between the many opposing charge instances (e^- and p^+) within a system (a.k.a. **Coulomb's Law**).

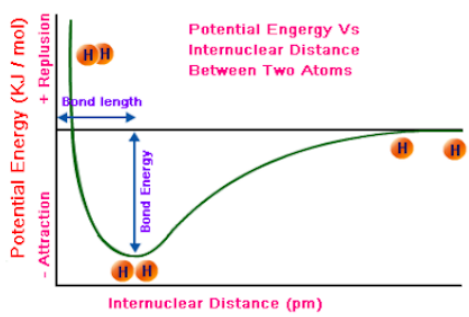

Figure 2: Potential Energy

Intramolecular bonds between atoms always arrange themselves at specific distance from one another that occur at the lowest PE possible. PE is the energy associated with the bonds between the atoms.

For instance, when the internuclear distance is **infinitely large**, there is no PE within the interaction betwen the atoms because there is virtually no force between them.

Once atoms approach one another, they will reach an equlibrium point where the repulsive (*like charge interactions*) and attractive forces (*opposing charge interactions*) across atoms equal one another. The PE at this equilibrium point is called the **bond energy** and results in a release of energy, analagous to two magnets snapping and releasing heat on impact. At this point, we would need to expend or input energy in order to break the bond between the atoms, thus the PE is indicated as negative, analagous to two magnets being pulled apart by an input of force.

At an even closer distance, the PE will increase because the repulsive forces overcome the attractive forces, thus resulting in a higher PE state than at the equilibrium point, analogous to two rubber balls being pressed against

each other.

Remember, **breaking a bond will always require an INPUT of energy**, thus the change in PE from the bonded to the unbonded state is negative. Conversely, **forming a bond will always result in an OUTPUT of energy**, thus the change in PE from the unbonded to the bonded state is positive.

Within a closed system, the total energy of the system is equal to the sum of the PE and KE of the system. This is represented mathematically as:

$$E_{total} = KE + PE$$

If **PE drops** (e.g., two atoms snap together to form a bond), the system must gain KE. This is why exothermic reactions release heat—the potential energy "lost" by the atoms settling into a stable bond is converted into the kinetic energy of the molecules vibrating and moving faster.

If **KE drops** (e.g., a molecule slows down as it flies away from an attractive neighbor), that energy hasn't vanished; it has been investing into PE as the molecule climbs out of a potential well.

1.2 Heat

Let's revisit endo- and exothermicity.

An **ENDOTHERMIC REACTION**, is a reaction that requires the input of heat in order to occur. Take the melting of ice for example. An **EXOTHERMIC REACTION**, is a reaction that releases heat in order to occur. Take the combustion of gasoline for example.

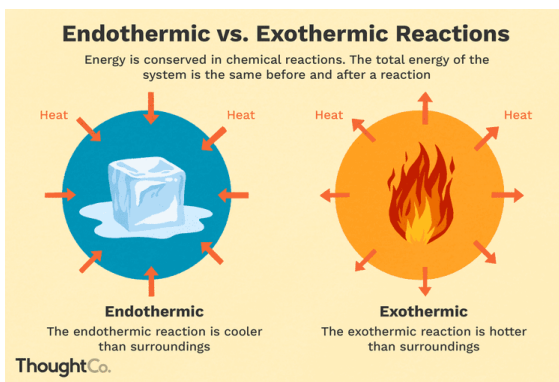

Figure 3: Endothermic and Exothermic Reactions

Definition 1.1: Example

It's important to remember that "**hot**" and "**cold**" describe how much particles are moving. Hot substances have particles that vibrate and move more, while cold substances have particles that vibrate less. Temperature reflects the average kinetic energy of the particles.

When two systems with differing temperatures come into contact, the system with the lower temperature will absorb heat from the system with the higher temperature. This is called **HEAT TRANSFER**.

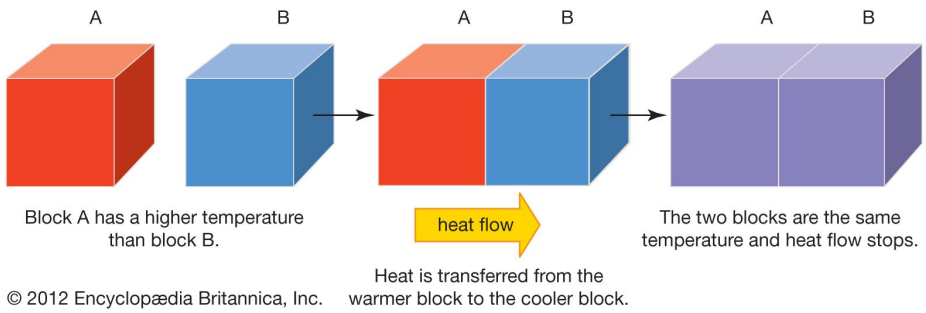

Figure 4: Heat Transfer

Chemists represent the transfer of heat with this equation:

$$q = mC\Delta T$$

where q is the amount of heat transferred, m is the mass of the system, C is the **specific heat capacity**, and ΔT is the change in temperature.

Specific heat capacity is a measure of the energy required to raise the temperature of 1g of a substance by 1 K. For instance, the specific heat capacity of water is $4.184 J\, g^{-1} K^{-1}$. In other words, it takes 4.184 joules to raise the temperature of 1g of water by 1K.

As an example, let's find out how many joules it takes to raise the temperature of 43g of water by 45K.

$$q = (43g)(4.184 J\, g^{-1} K^{-1})(45K) = 8096.04 J$$

While joules are the most common unit for energy in the context of chemistry, another common energy unit is **calories**, which is equal to 4.184 J. With this in mind, the specific heat capacity of water in $cal\, g^{-1} K^{-1}$ is just 1.

Definition 1.2: Example

The calorie figures often shown in nutrition labels are written in Cal or kcal, which is 1000 calories.

Let's take a look at another example. **You place 560g of Au at 100°C into 560mL of water at 20°C. What would be the final temperature of the water?** $C_{H_2O} = 4.184 J\, g^{-1} K^{-1}$, $C_{Au} = 0.13 J\, g^{-1} K^{-1}$.

Recall that energy cannot be created nor destroyed in a closed system. Thus the total net energy change must be zero.

$$q_{lost} + q_{gained} = 0$$

which we can rewrite as:

$$q_{gained} = -q_{lost}$$

Heat is transferred from the Au to the water, thus water gains energy while Au loses energy. The T_{final} of Au would be the same as the T_{final} of the water. Thus, the equation becomes:

$$(560g)(4.184Jg^{-1}K^{-1})(T_{final} - 20) = -(560g)(0.13Jg^{-1}K^{-1})(T_{final} - 100)$$

Solving for T_{final} brings us to $T_{final} = 22.41°C$

Okay, now let's move onto phase transitions, namely of water. We call the heat required to transition water from a solid to a liquid state as the **heat of fusion**, and the heat required to transition water from a liquid to a gas state as the **heat of vaporization**. The signs of these variables depends on whether you are evaporating or condensing (*for vaporization*) and whether you are melting or freezing (*for fusion*).

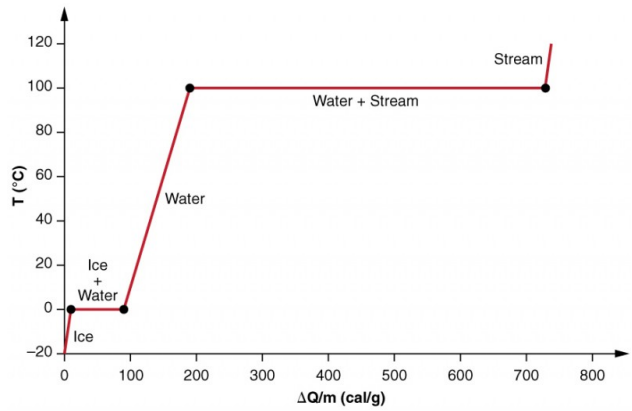

Figure 5: Heating Curve of Water

As you can see from the graph above, the temperature does not increase as vaporization occurs. During this process, energy is spent to raise the PE of the IMFs between water molecules rather than their KE (a.k.a. what temperature measures). In other words, at 100°C, the molecules in liquid water are moving with the same average KE as they will be once they become steam at 100°C. The "vibrational speed" and "collisional energy" of liquid state water molecules are already up to par as that of the gas phase, but the molecules are still stuck in the liquid phase because of the deep PE wells created by hydrogen bonding IMFs. The heat of vaporization is the energy required to seperate the molecules from neighboring molecules (effectively break the IMFs and raising PE) in order to evaporate into a gas phase.

This same logic applies to the heat of fusion.

You may also notice that the energy magnitude for the heat of vaporization is much more than that of fusion. This is because during fusion, you are only adding enough energy to **disrupt the rigid, ordered structure of the crystal lattice** whereas during vaporization, you are adding enough energy to overcome the IMFs between water molecules **completely**. The scale difference between the two is enormous as a result.

The experimental science of measuring the amount of heat (q) exhcanged during a chemical reaction, physical chnage, or between two objects at different temperatures is called **CALORIMETRY**. It is the practical application of the **First Law of Thermodynamics (Conservation of Energy)**.

1.3 Bond Energy

Enthalpy (H) is a thermodynamic property that represents the total heat content of a system. It accounts for both the internal energy of the system and the energy required to "make room" for it by pushing back the surrounding atmophere.

While you cannot measure the absolute enthalpy of a system, you can measure the change in enthalpy (ΔH) during a process, which is why this term is so central to chemistry.

Let us revisit the bond energy graph from the *3.1.1 Potential and Kinetic Energy*.

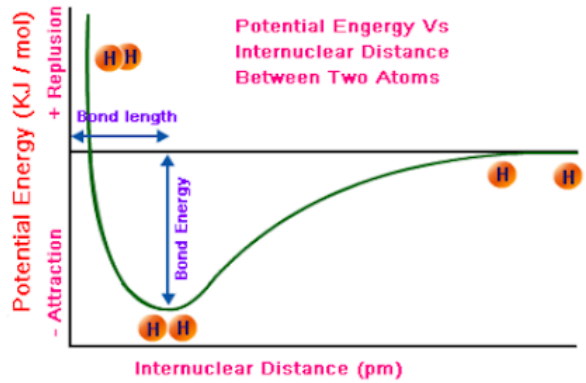

Figure 6: Potential Energy Diagram

The formation of bonds releases energy and the breakdown of bonds requires energy. All chemical reactions are are the breakdown of the initial bonds of reactants and the formation of new bonds of products. ΔH is used to denote whether a reaction results in the net input of heat (**endothermic**) or the net release of heat (**exothermic**)

Take the combustion of hydrogen to produce water for example.

$$2\,H_2 + O_2 \longrightarrow 2\,H_2O$$

Here is a graph of the enthalpy change during the reaction.

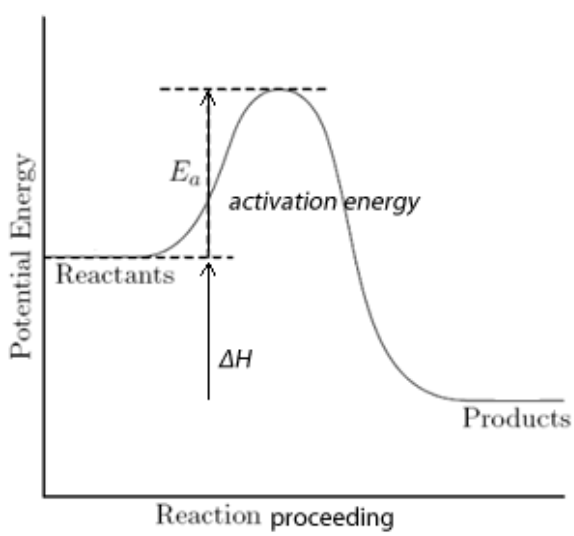

Figure 7: Enthalpy of Combustion

Let's first start off by imagining the reaction occuring in front of you. When you light a balloon of H_2, you'd see it light up and explode. The fact that heat was released as a result of the reaction should tell you that the reaction was exothermic.

What we saw was that the original reactants (the H_2 within the balloon and the O_2 in the air) were transformed into the products (H_2O) as a result of an inital input of energy (the lighter). We call this initial input of energy as the **activation energy**, the jumpstart in energy the reaction needs to carry out the breakdown of reactant bonds before

reforming product bonds.

The fundamental driving force of chemistry is **the pursuit of the lowest PE state**. Thus the bonds of the free atoms (H, H, H, H, O, O) would interact to form 2 H_2O. And since the enthalpies of the bonds exist at a lower PE than the original bonds, the net release of heat equal to the net difference in enthalpy occurs (the explosion).

Definition 1.3: Example

It may be worth noting that a small spark is not enough to fulfill the activation energy of all H_2 and O_2 molecules at once. Instead, the energy release from one spark is sufficient to form one H_2O molecule. The release of energy from the formation this H_2O is then used as the E_a of neighboring H_2 and O_2 molecules. This domino effect occurs at a rapid rate to carry out the reaction for the entire balloon.

We can represent the enthalpy change quantitatively as well. These are the average bond enthalpy values for several commonly occurring bonds.

Bond	H–H	O–O	H–O	C–C	C=C	C≡C	C–O	O=O
$H_{average}$ (kJ/mol)	436	146	463	348	612	838	360	498

We can calculate the enthalpy change for the breakdown and formation of each bond as follows:

Break 2 moles of H–H → (2)(+436 $kJ\,mol^{-1}$)
Break 1 moles of O=O → (1)(+498 $kJ\,mol^{-1}$)
Make 4 moles of O–H → (4)(−463 $kJ\,mol^{-1}$)

Thus $\Delta H = -482\,kJ\,mol^{-1}$ for the reaction.

Definition 1.4: Example

It is also important to remember that E_a is rarely the energy required to break ALL the reactant bonds. **E_a is only the energy needed to break key reactant bonds just enough for the reaction to occur**. For instance the E_a of water formation is only 200 $kJ\,mol^{-1}$, not the entire 1370 $kJ\,mol^{-1}$ required to break all reactant bonds.

Let's take a look at another example: the autoionization of water.

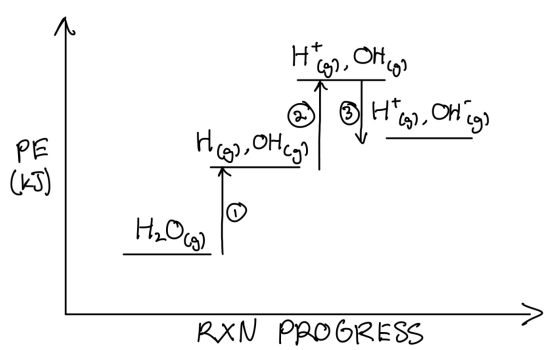

Figure 8: Autoionization of Water Enthalpy

Here, we see a net rise in PE, making the overall reaction endothermic ($\Delta H > 0$). The reaction is seperated into three distinct steps

(1) Break 1 mole of O–H → (1)(+467 $kJ\,mol^{-1}$)

(2) Pull e^- from 1 mole of H ("ionization energy") → (1)(+1312 $kJ\,mol^{-1}$)

(3) Add e^- to 1 mole of O ("electron affinity") → (1)(−180 $kJ\,mol^{-1}$)

Thus $\Delta H = +1599\,kJ\,mol^{-1}$ for the reaction.

If two or more chemical equations are added to give another chemical equation, the corresponding enthalpies of each reaction can be added to obtain the enthalpy of the combined reaction. This is called **Hess's Law**. Enthalpies are **state functions**, meaning it does not matter what path you take, the altitude difference is the same.

Lets take a look at the reaction of iron oxide with carbon.

This is the chemical equation of the formation of iron oxide. The reaction is exothermic.

$$2\,Fe(s) + \tfrac{3}{2}O_2(g) \longrightarrow Fe_2O_3(s) \qquad \Delta H = -824\,kJ\,mol^{-1}$$

Thus the opposite reaction (*iron oxide deomposition*) is endothermic. We'll call this reaction **Rxn A**.

$$\text{(Rxn A) } Fe_2O_3(s) \longrightarrow 2\,Fe(s) + \tfrac{3}{2}O_2(g) \qquad \Delta H_A = +824\,kJ\,mol^{-1}$$

Let's also take a look at the combustion of carbon. We'll call this reaction **Rxn B**.

$$\text{(Rxn B). } C(s) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H_B = -394\,kJ\,mol^{-1}$$

When we combine 2 Rxn A with 3 Rxn B, we get the following reaction of iron oxide with carbon. We'll call this reaction **Rxn C**.

$$\text{(Rxn C) } 2\,Fe_2O_3(s) + 3\,C(s) \longrightarrow 4\,Fe(s) + 3\,CO_2(g)$$

Based on Hess's Law, the sum of the enthalpy changes of 2 Rxn A and 3 Rxn B is equal to the enthalpy change of the combined reaction.

$$\Delta H_C = 2\Delta H_A + 3\Delta H_B = +466\,kJ\,mol^{-1}$$

Thus, it would take +466 $kJ\,mol^{-1}$ to deoxidize 2 moles of iron oxide or +233 $kJ\,mol^{-1}$ to deoxidize 1 mole of iron oxide. Making this process much more favorable compared to the +824 $kJ\,mol^{-1}$ net enthalpy change in the absence of carbon.

This chemical reaction is process of smelting raw iron ore into pure iron.

Let's look at another example of the formation of $H_2O(g)$ from $H_2O(l)$ at 25°C.

$$H_2O(l) \longrightarrow H_2O(g) \qquad \Delta H_{Rxn} = ?$$

$$\text{(Rxn A) } H_2O(l) \longrightarrow H_2(g) + \tfrac{1}{2}O_2(g) \qquad \Delta H_A = +286\,kJ\,mol^{-1}$$

$$\text{(Rxn B) } H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O(g) \qquad \Delta H_B = -242\,kJ\,mol^{-1}$$

Thus

$$\Delta H_{Rxn} = \Delta H_A + \Delta H_B = +44\,kJ\,mol^{-1}$$

or

$$\Delta H_{Rxn} = \sum \Delta H_f(products) - \sum \Delta H_f(reactants)$$

In other words, this formula is equivalent to **undoing the formation of the reactants** and then **doing the formation of the products**.

2 Entropy and Gibbs Free Energy

2.1 Defining Entropy

Enthalpy is just one factor in predicting whether a reaction will be favorable or unfavorable. For instance, the dissociation of ammonium chloride into its cooresponding ions is endothermic, yet it goes to completion.

$$NH_4Cl \longrightarrow NH_4^+ + Cl^- \qquad \Delta H_{Rxn} = +14.8\, kJ\, mol^{-1}$$

What about the tendency for gas to expand into a vacuum? Although this isn't a reaction, it occurs in only one direction and $\Delta H = 0$, thus it cannot be explained by enthalpy alone.

There is a natural tendency for energy and matter to disperse. In chemistry terms, this means that systems tend to react towards a state with high **entropy**. This is also known as **the second law of thermodynamics**.

Introductory chemistry textbooks often describe entropy as *disorder*. However, this is a very vague definition. Instead, entropy is more accurately defined as a measure of how many different ways (**microstates**) you can achieve a specific result (**macrostate**).

Take the sum of rolling two dice for example. There are a total of 36 different combinations of rolling 2 dice, but the statistical likelihood of of sum outcomes is concentrated around 7. The outcome of the sum being 7 is the macrostate in this case, and the different combinations of possible dice rolls (6-1, 5-2, 4-3, 3-4, 2-5, 1-6) are the microstates. It is much less likely to be at a low entropy macrostate (*like the sum of 2*) than a high entropy macrostate (*like the sum of 7*) because of the number of valid microstates for each macrostate.

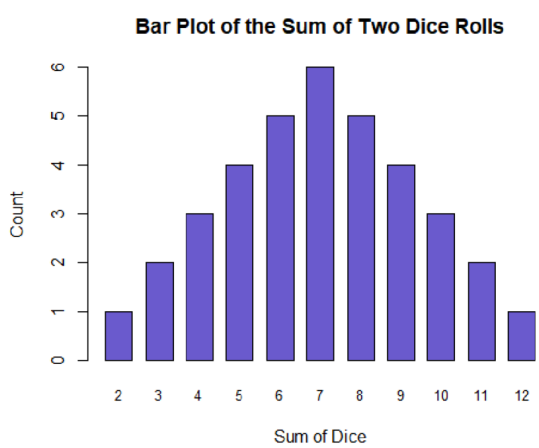

Figure 9: Dice Roll Probability Distribution

Now let's think about this in the context of gas molecules within a closed container at 25°C. Intuitively, we can expect the gas to spontaneously disperse throughout the container volume. Just as rolling a sum of 7 on two dice is the most likely macrostate because of its high number of possible microstates (*number combinations*), the gas utilizing the entire container volume is the most likely macrostate because of its higher number of possible microstates (*positional combinations*). The low entropy macrostate counterpart is if all gas molecules concentrate into one corner of the container. There are much less possible microstates in which the gas molecules can arrange themselves in when restricted to a small corner compared to the entire container.

The following figure shows this concept nicely but at a smaller scale.

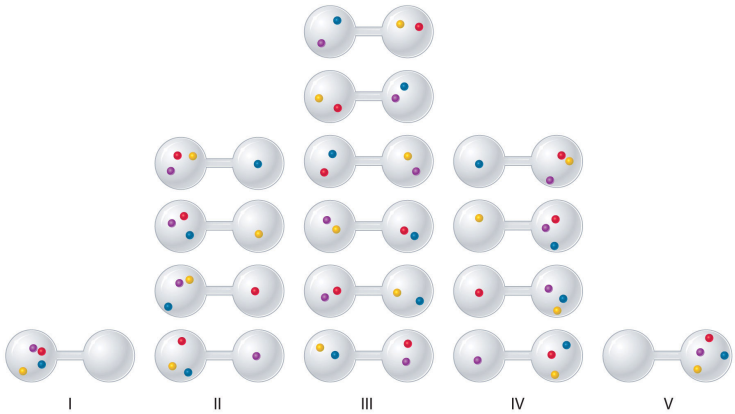

Figure 10: Gas Molecule Position Microstates

So far, we've only talked about the coordinate positions of the gas molecules. In reality, there are also many other characteristics that we can include to expand the number of possible microstates there are for a given macrostate: rotational position, vibrational motion, translational motion, magnitude of KE, etc.

Dimensions tied to KE are consequently tied to temperature.

Like the gas molecule coordinate positions, there are more possible microstates in which the distribution of T in the form of KE is more widely distributed across the entire container volume compared to being concentrated in one corner, hence the tendency for heat to transfer into cooler surroundings.

When we say that the gas molecules are at 25˚C, we're actually referring to the average KE of the gas molecules, NOT all gas molecules being at the KE equivalent of 25˚C simultaneously. At a random time instance, one gas molecule could be at the KE equivalent of 24.99˚C, and another at 25.01˚C. With that being said, there are many more possible microstates in which the KE magnitudes of the gas molecules are within a wider range compared to all being at that of 25˚C.

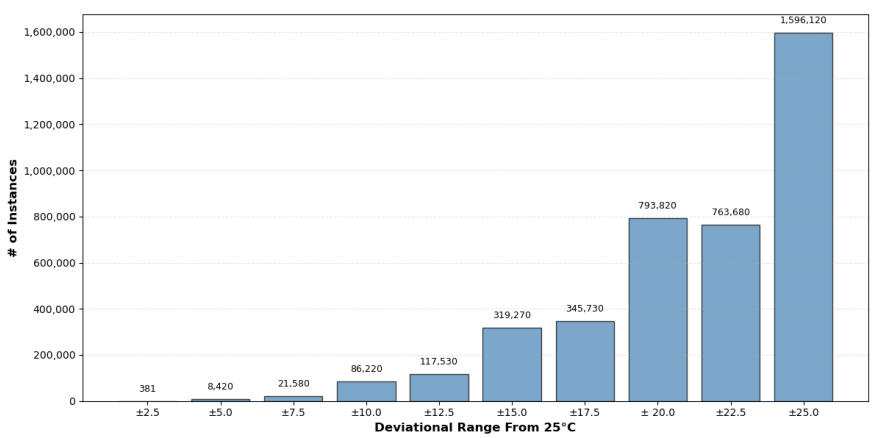

Figure 11: Gas Kinetic Energy Distribution

The graph above represents the counts of possible 5 number permutations that sum to 125 (and thus average out to 25) seperated by the maximum deviation from 25 that was allowed. The possible permutation ranges were also constrained to 0-50 inclusive for the sake of simplicity. As you can see, even in an extremely simplified model like this, there is a much larger number of possible microstates where the temperature ranges are more widely distributed than when all gas molecules are at the same KE such as the permutation (25, 25, 25, 25, 25). Thus there is a tendency

for heat to be distributed more randomly through out gas molecules although the average KE is still 25˚C. This is the **Maxwell-Boltzmann Distribution** that we've visited *1.2.1 Kinetic Molecular Therory and Ideal Gas Laws*.

With that being said, entropy (S) can be defined through the equation

$$S = k_B \ln(W)$$

where W represents the number of energetically equivalent microstates of a system and k_B is the Boltzmann's constant represented as

$$k_B = \frac{R}{N_A} = \frac{8.314 J\,K^{-1}\,mol^{-1}}{6.022 \times 10^{23} units} = 1.38 \times 10^{-23}\,J\,K^{-1}$$

where R is the ideal gas constant in $J\,K^{-1}\,mol^{-1}$ and N_A is Avogadro's number, which represents the number of units per mole. k_B tell us how much thermal energy is contained in a single particle for every degree of temperature in a system.

And from our discussion of entropy, we can see that the change in number of accessible microstates is exponentially related to the change in the number of particles, thus the proportionality of entropy is to $ln(W)$ and not simply W. This relationship will be brought back to in *3.2.3 Gibbs Free Energy and Equilibrium Constants*.

In more general terms, entropy increase ($\Delta S > 0$) as (1) temperature increases, (2) a reaction results in a net increase in number of molecules, and (3) a phase change towards more unstable phases occurs. When presented with more nuanced cases, always revert back to whether the change or reaction will increase or decrease the number of possible microstates.

2.2 Changes in Entropy and Gibbs Free Energy

The **third law of thermodynamics** states that as the T of a system approaches absolute zero (0 Kelvin), its entropy approaches a constant minimum value, which is 0 for a perfect, pure crystal. This is critical because it ables us to quantify the absolute entropy of a system, molecule, mole of a substance, etc. Recall that we could not do the same with enthalpy because we could not quantify the absolute enthalpy of a system due to the absence of an absolute reference point (*we could only calculate its change*).

We measure entropy in the units of **Joules per Kelvin (J/K)**. While this unit may be difficult to wrap your head around, it is easier when compared to a similar unit like grams per liter (g/L) or density. Just as how density is a descriptive value of a substance of how much mass it would have at a certain volume, entropy is a descriptive value of how large the range of microstates it has access to (quantified as energy) for a certain temperature.

It's also important to keep in mind that **the entropy of a system (*the ratio J/K*) increases with temperature**, but at a decreasing rate. Conceptually, we can imagine inputing the same amount of energy into a system of boiling water vs ice. The boiling water is already at a state of high entropy thus a 1 K increase in temperature does a lot less percent change to the system compared to a 1 K increase in temperature for ice, which is at a state of low entropy. We express this mathematically as

$$\frac{dS}{dT} = \frac{C_p}{T}$$

This equation says that the rate at which entropy increases with temperature is proportional to the heat capacity of the system and inversely proportional to the temperature. We explained why entropy is inversely proportional to temperature above. C_p is proportional to the rate of entropy change because a higher C_p indicates a higher input of energy per Kelvin increased in temperature. For instance, you'd need to input a lot more energy to raise water by 1 K compared to Cu. Because you are pumping in so many Joules just to move the needle by 1 K for water, you are

creating a massive amount of new molecular "chaos", whereas you aren't putting in as much energy in Cu for a 1 K uptick, thus less new "disorder".

Let's discuss the case of increasing the temperature of water from 25°C to 50°C in the context of enthalpy and entropy.

Enthalpy is simple calorimetry as we've learned previously. ΔH is simply the the products of mol_{H_2O}, C_{H_2O}, and ΔT.

In order to quantify this energy input through entropy gain, we must consider (1) the the increasing entropy of water as T increases and (2) the decreasing rate at which entropy increases as T increase. Mathematically, this is summarized as

$$\frac{dS}{dT} > 0, \frac{d^2S}{dT^2} < 0$$

The area under the curve of a line graph presenting S as a function of T, from the initial T to the final T, represents the access to a greater number of microstates and/or a higher entropy. The overall shape of the line would be similar to $f(x) = ln(x)$.

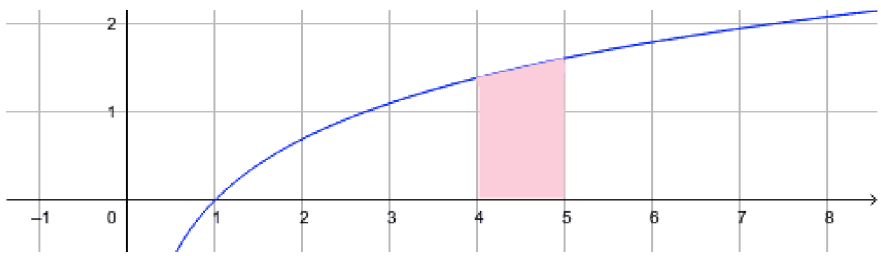

Figure 12: Entropy vs Temperature

Definition 2.1: Example

When we get really nuanced with the definitions of entropy and enthalpy, we realize that enthalpy is a measure of temperature dependent entropy characteristics in a way. The limitations of enthalpy is that you cannot quantify the change in entropy dependent on positional coordinates, which stretch outside the scope of this course.

Okay, now let's revisit the **second law of thermodynamics** which states that the total entropy of an isolated system always increases over time. Our universe is essentially an isolated system (*albeit a very large one*), thus we can represent this law as the equation

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{environment} > 0$$

Let's look at a scenario where we melt 10 moles of water. The energy inputted into the system would be the enthalpy of fusion, and since the temperature of the system is static during phase changes, we do not need any integral calculations. Thus the entropy change of the ice cube (the system) would be

$$\Delta S_{system} = \frac{\Delta H_{fusion}}{T_{system}} = \frac{(6.02\,kJ\,mol^{-1})(10\,mol)}{273\,K} = +220J\,K^{-1}$$

As for the surroundings, the temperature does increase (*in theory*), but the shift is negligible due to the surroundings being quite literally the rest of the universe. As a result, we keep $T_{surroundings}$ at 298K. In most chemistry problems,

the "surroundings" are modeled as a **thermal reservoir**. By definition, a reservoir is so large that its heat capacity is effectively infinite in comparison to the system.

With this in mind, the cooresponding entropy change of the surroundings would be

$$\Delta S_{surroundings} = \frac{-\Delta H_{fusion}}{T_{surroundings}} = \frac{-(6.02\,kJ\,mol^{-1})(10\,mol)}{298\,K} = -202J\,K^{-1}$$

Consequently, $S_{universe}$ would then be

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} = 220J\,K^{-1} + (-202J\,K^{-1}) = +18J\,K^{-1}$$

Conceptually, this means that (1) the entropy of the system increased due to the input of heat into the system and (2) the entropy of the surroundings decreased due to the transfer of heat from the surroundings to the system (*but to a lesser degree because the marginal difference in the context of the greater temperature of the surroundings is smaller as we've explained previously*).

Let's algebraically manipulate the equation for $S_{universe}$.

$$\Delta S_{uni} = \Delta S_{surr} + \Delta S_{sys} > 0$$

$$\Delta S_{uni} = \frac{-\Delta H_{sys}}{T_{surr}} + \Delta S_{sys} > 0$$

$$T_{surr}\Delta S_{uni} = -\Delta H_{sys} + T_{surr}\Delta S_{sys} > 0$$

$$-T_{surr}\Delta S_{uni} = \Delta H_{sys} - T_{surr}\Delta S_{sys} < 0$$

We have just defined the **Gibbs free energy** equation where $\Delta G = -T_{surr}\Delta S_{uni}$.

$$\Delta G = \Delta H_{sys} - T_{surr}\Delta S_{sys} < 0$$

Gibbs free energy is another way to conveying whether the total entropy of the universe is increasing or not (*a net spontenaity score*). It acts as a balancing act between two universal desires: (1) to maximize the entropy of the universe (2) to minimize the enthalpy of the system. If a reaction fulfills both conditions, the reaction is **spontaneous**. If a reaction does not fulfill both conditions, the reaction is **non-spontaneous**. If a reaction fulfills only one, the reaction's spontenaity is determined by which force is greater.

$\Delta H°$	$\Delta S°$	T	$\Delta G°$	Reaction Favorability
−	+	ANY	−	Product favored (→)
−	−	HIGH	+	Reactant favored (←)
−	−	LOW	−	Product favored (→)
+	+	HIGH	−	Product favored (→)
+	+	LOW	+	Reactant favored (←)
+	−	ANY	+	Reactant favored (←)

ΔG_{rxn} measures the difference between the free energies of the reactants and products when all components of the reaction are present at standard-state conditions. ΔG_{rxn} can be either calculated from cooresponding ΔH_{rxn} and ΔS_{rxn} values or from the the Gibbs free energies of formation (ΔG_f). Like both enthalpy and entropy are state functions, ΔG_{rxn} is also a state function.

$$\Delta G_{rxn} = \Delta G_f(products) - \Delta G_f(reactants)$$

An easy way to conceptualize the Gibbs free energy equation is to treat ΔS as a scaling factor for the effect of temperature on free energy. Let's take a look at the reaction between barium hydroxide and ammonium chloride.

$$Ba(OH)_2 \cdot 8\,H_2O(s) + 2\,NH_4Cl(s) \longrightarrow BaCl_2(aq) + 2\,NH_3(aq) + 10\,H_2O(l)$$

This reaction is highly endothermic ($\Delta H = 62\,kJ\,mol^{-1}$). However, just by first glance at the states of the reactants vs products, we can see that this reaction raises entropy tremendously ($\Delta S = 368\,J\,mol^{-1}\,K^{-1}$). Thus the free energy provided by the product entropy is much greater than that of the reactant entropy at the same temperature because the products have an access to a greater number of possible microstates due to several factors (*number of moles of molecules, phases, etc*). The magnitude of this "entropic energy" gain at 298K is greater than the magnitude of the enthalpy increase the reaction provides, thus the reaction is favored.

Let's imagine water states at a theoretical temperature of 0 Kelvin. We already know that liquid water will freeze at absolute zero but why? First, let us take a look at the freezing phase change reaction.

$$H_2O(l) \longrightarrow H_2O(s)$$

The process is exothermic thus the process is enthalpically favorable. However, the entropy falls as the phase change from liquid to solid occurs. Yet this entropy drop does not matter in the context of absolute zero because the free energy (*or access to microstates*) of liquid water and solid water would be the same. *It is counterintuitive to conceptualize liquid water at absolute zero because water as we know it doesn't exist in the liquid state at absolute zero. But this phenomenon is exactly what we are trying to explain.* At absolute zero, entropic free energy is zero, thus the only factor that matters is enthalpy.

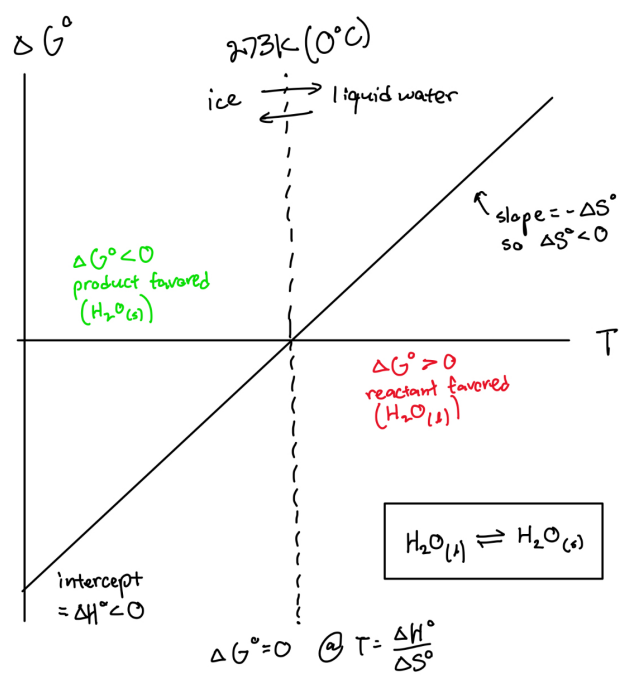

Figure 13: Gibbs Free Energy of Water

At higher temperatures, the free energy bound the entropy of liquid state of water is always greater than that of the solid state. However, until this difference in free energy bound to entropy surpasses the enthalpic change that occurs during freezing (*which approaches as T increases*), the frozen state of water is favored. It is at 273K that we see the phase change from solid to liquid occur. This is where free energy bound to entropy drops too much during a freezing reaction to push Gibbs free energy into the positives, thus the freezing phase change reaction is not favored and the reverse is favored.

FINAL TAKEAWAY: The Gibbs free energy equation allows us to gauge the favorability of a reaction by comparing the free energies provided by the enthalpy changes across the reactants and products and the free energies provided by the entropy change across the reactants and products. Entropy changes at a higher surrounding T will magnify the effects of entropy differences on free energy. This is because at higher temperatures, the system has more thermal energy to distribute among its avilable microstates, making the free energy "reward" for an already greater number of microstates (high entropy) much larger vs a lower number of microstates (low entropy). The universe always wants to minimize free energy or the potential to do spontaneous work.

2.3 Gibbs Free Energy and Equilibrium Constants

We have established so far that if a reaction's ΔG is negative, it is favorable, and if it is positive, it is unfavorable. However, from out previous chapter on change and equilibrium constants (K), we know that just because a reaction is favorable, it does not mean that it goes to completion. Many reactions find an equlibrium point where some reactants or products are still leftover although their presence is not favored in the context of ΔG.

There must thus be a quantitatively defined relationship between ΔG and K. Here, we introduce the **Van 't Hoff Isotherm** equation.

$$\Delta G^\circ = -RT \ln K$$

Based on our previous establishment of $S = k_B \ln(W)$, we can see that entropy has a logarithmic relationship with the number of possible microstates (*which shift exponentially to the number molecules in a system*). Since K relates to the concentrations (*or the amounts*) of reactants and products, the accessible number of microstates change exponentially when Q shifts. Thus since ΔG incorporates ΔS into its value, the relationship between K and ΔG is logarithmic. Consequently, a linear change in ΔG leads to an exponential change in K.

Also recognize the relationship between ΔG and T. A change in T will have an effect on the magnitude of ΔG but also shift the value of K based on the ΔH of the reaction. The T at which K is exactly 1 would be when neither side of the reaction is inherently favorable and thus there an equal part reactant and product ratio.

Under non-standard conditions in the context of the current reactant product ratio Q. We would represent the total potential free energy left in the system by

$$\Delta G = \Delta G^\circ + RT \ln Q$$

or equivalently

$$\Delta G = -RT \ln \frac{K}{Q}$$

When $K = Q$, we have achieved equilibrium, thus there is no more potential free energy left to do work. If $K > Q$ or $K < Q$, the reaction still has free energy left to do work. This these cases the sign of ΔG represent the direction of the reaction that can perform work.

3 Examples and Applications

3.1 Work

The free energy released from a system can be converted to work and used to exert influence on an external body. While reactions that are endothermic but entropy increasing enough to be favorable will be able to fulfill this purpose, it is more common to see reactions that are exothermic to be the source of work. In many cases, these exothermic reactions tend to decrease entropy along the process, thus the free energy that we can use to drive external processes will be less than the total heat released by the reaction.

This is what Gibbs free energy is used for in a practical sense. Let's take a look at the Gibbs free energy equation again.

$$\Delta G = \Delta H - T\Delta S$$

For $\Delta H < 0$, ΔH can be converted into work after "the entropy cost" is paid for.

For $\Delta S < 0$, ΔS must be compensated for in order to satisfy the second law of thermodynamics ("*the entropy cost*").

For $\Delta S > 0$, ΔS can also be converted into work.

Let's go through a collective example of how a system's change in entropy and enthalpy can be used to drive work onto an external system.

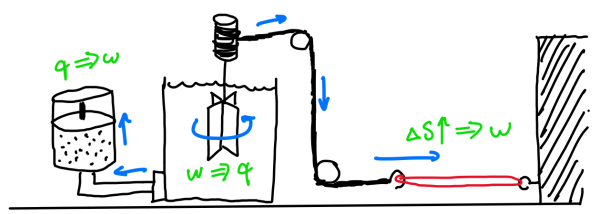

Figure 14: Entropy to Work Conversion

The figure above shows an initial state of a system. In this initial state, the red rubber band is in its stretched, low entropy position because there are fewer microstates in which the chains can be straight compared to its relaxed state where the chains are free to access a larger range of microstates because they aren't confined by the requirement to be straight.

ENTROPY TO WORK: As the rubber band contracts, the free energy provided by this rubber band system's entropy rise is converted into work that pulls the cable it is connected to.

WORK TO HEAT: As the work transferred from the rubber band enters the cable-paddle system (rotating the paddle), it immediately leaves the system as heat is inputted into the water system via friction from the paddle.

HEAT TO ENTROPY TO WORK: This heat energy is transferred onto a gas system within a piston chamber to push the piston up via the gas system's entropy increase due to an increase in volume.

Definition 3.1: Example

It is important to note that **the process of heat conversion to work is never a perfect conversion**. This is because the quality of energy from work is different from that of heat. Heat is the random vibration of molecules within a system, thus making it inherently high entropy. Work, on the other hand, is the organized

motion of molecules within a system, thus making it low entropy. **If heat to work conversion is perfect, the universe will lose entropy in the process** (*we know from the second law of thermodynamics that this is impossible*), thus a portion of heat is released into the relatively lower temperature surrounding to compensate for the loss of entropy from the heat to work conversion process.

Internal energy (E) of a system is the sum of all kinetic and potential energies of all the system's components.

$$\Delta E = q + w$$

Energy gained or lost by the system is equal to the energy lost of gained by the surroundings.

To help us better understand this concept, imagine the transfer of heat energy into two gas systems to T_{final} from $T_{initial}$. We restrain one system to constant volume and the other to constant pressure.

CONSTANT VOLUME: The system is restricted from doing any expansion work to its surroundings, thus 100% of the heat you add goes directly into increasing the E of the system (*raising T*).

CONSTANT PRESSSURE: This system, on the other hand, is free to expand. Thus for the same T increase as the former system, you would need to add extra heat energy (*to raise T to* T_final) equivalent to the energy transferred to the surroundings from the expansion work it will do as volume increases.

It would take more heat energy to raise the temperature of the system with constant pressure than the system with constant volume by the same temperature increase.

3.2 Electrochemical Cells

An electrochemical cell is a device that bridges the gap between chemical energy and electrical energy, effectively allowing us to use the free energy drop of a spontaneous chemical reaction to generate electrical energy for practical purposes. It works by physically separating a redox (reduction-oxidation; e^- *transfer reactions*) reaction so that e^-s are forced to travel through an external circuit rather than transferring directly between atoms within a solution.

Before introducing electrochemical cells, we first need to understand electrode potentials. **Electrode potential** is a measure of the tendency of a metal to either "shed" its atoms into a solution as ions or "pull" ions from the solution to become part of the metal.

A high electrode potential means the metal is more likely to pull ions from the solution to become part of the metal. This is the reduction of metal ions into neutral metal atoms.

$$M^+(aq) + e^- \longrightarrow M(s)$$

A low electrode potential means the metal is more likely to shed ions into a solution from the metal. This is the oxidation of neutral metal atoms into metal ions.

$$M(s) \longrightarrow M^+(aq) + e^-$$

You cannot measure the potential of a single metal electrode in isolation. You always need at least one more metal electrode to compare it to. To solve this, chemists have designated a universal reference point to hydrogen called the **Standard Hydrogen Electrode (SHE)**, which is arbitrarily assigned a value of 0.00 V. Here are the electrode potentials of some common metals:

Potential	Reduction Half-Reaction
−3.05 V	$Li^+(aq) + e^- \longrightarrow Li(s)$
−2.71 V	$Na^+(aq) + e^- \longrightarrow Na(s)$
−2.37 V	$Mg^{2+}(aq) + 2e^- \longrightarrow Mg(s)$
−1.66 V	$Al^{3+}(aq) + 3e^- \longrightarrow Al(s)$
−0.76 V	$Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)$
−0.28 V	$Ni^{2+}(aq) + 2e^- \longrightarrow Ni(s)$
0.00 V	$2H^+(aq) + 2e^- \longrightarrow H_2(g)$
+0.34 V	$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$
+0.80 V	$Ag^+(aq) + e^- \longrightarrow Ag(s)$
+1.06 V	$Br_2(l) + 2e^- \longrightarrow 2Br^-(aq)$
+1.23 V	$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$
+1.36 V	$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$
+2.87 V	$F_2(g) + 2e^- \longrightarrow 2F^-(aq)$

We assigning the units of V or J/C (joules per coulomb or energy per charge) because the potential tells us how much energy each e^- possesses as it spontaneously travels from a low potential electrode to a high potential electrode.

With that being said, if you were to place Zn metal powder into a solution of Cu ions, a spontaneous redox reaction would occur where the solid Zn metal would eventually turn into solid Cu.

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

Since this redox reaction is spontaneous ($\Delta G < 0$), free energy is released. A **galvanic cell** (*a type of electrochemical cell*) is a set up that allows us to capture this free energy and use it as electrcity.

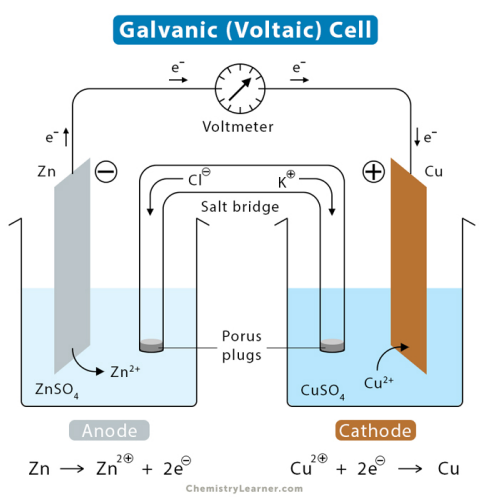

Figure 15: Galvanic Cell Diagram

In a galvanic cell set up, we call the low and high electrode potential metal the anode and cathode, respectively. In the above schematic, the anode is Zn and the cathode is Cu. Since the solutions of the two electrodes are physically seperated, the presence of a salt bridge allows each solution to maintain electrical neutrality and keep the redox

reaction moving forward rather than slow down and eventually stop due to the eventual build up of one-sided charge (positive ions on the anode from Zn^{2+}(aq) accumulation and negative ions on the cathode from ${SO_4}^{2-}$ presence with the decreaseing amounts of Cu^{2+}(aq)). The result is a system that captures the potential differnce between the two electrodes until the redox reaction is complete, at which point the anode is completely in its aqueous ion form and the cathod is completely in its solid form.

We can express the free energy gained from this system via the following equation:

$$\Delta G^\circ = -nFE^\circ$$

where n is the number of moles of e^-, F is the Faraday constant which measures the amount of charge per mole of e^- ($96485\, C\, mol^{-1}$), and E° is cell potential or the difference in potential between the anode and the cathode (in the above case it would be $0.34V - (-0.80V) = 1.14V$).

Let's go through a calculation for the galvanic cell above ($Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$).

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34V - (-0.80V) = 1.14V$$

$n = 2mol$ because two e^-s are being transferred across the cell. Thus

$$\Delta G^\circ = -(2\, mol)(96485\, C\, mol^{-1})(1.14\, J\, J^{-1}) = -219.98\, kJ$$

It's important to keep in mind that this free energy value is for every mole of the redox reaction that occurs, thus the same galvanic cell with Zn as its limiting reactant would have a free energy value of $-219.98\, kJ$ per mole of Zn.

Now we introduce a **concentration cell**, an electrochemical cell that uses the same metal for both the anode and cathode but at different solution concentrations.

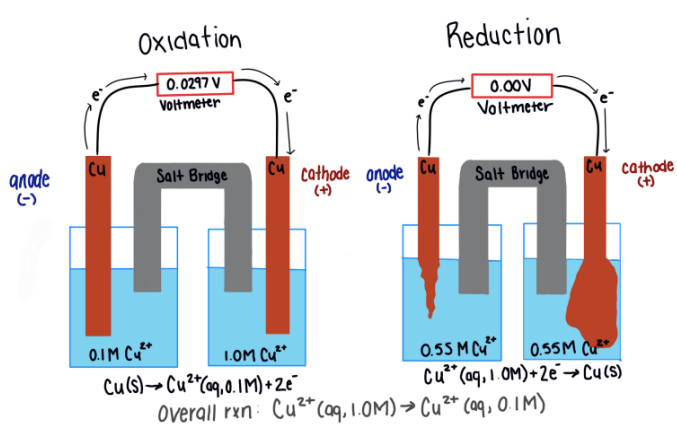

Figure 16: Concentration Cell Diagram

Let's take a look at the chemical equation for the redox reaction in the concentration cell.

$$Cu_{anode}(s) + {Cu_{cathode}}^{2+}(aq) \rightleftharpoons {Cu_{anode}}^{2+}(aq) + Cu_{cathode}(s)$$

The Eq for this reaction is $K = 1$, however

$$Q = \frac{[Cu^{2+}_{anode}]}{[Cu^{2+}_{cathode}]} < 1$$

Thus, the forward reaction if favored, raising the concentration of the anode solution at the expense of that of the cathode. Via the concentration differences, e^-s flow just like it did in the galvanic cell and we are able to capture the free energy of the redox reaction.

Quantiatively, we can measure the cell potential in volts using the following algebraic manipulations:

$$(STANDARD\,CONDITIONS) \quad \Delta G^\circ = -nFE^\circ = -RTln(K)$$

$$(NONSTANDARD\,CONDITIONS) \quad \Delta G = -nFE$$

$$\Delta G = \Delta G^\circ - (-RTln(Q))$$

$$-nFE = -nFE^\circ + RTln(Q)$$

$$E = E^\circ - \frac{RT}{nF} ln(Q)$$

Since E° is always 0 for concentration cells and $ln(x)$ $2.303 \times log_{10}(x)$, we can simplify the above equation to

$$E = -\frac{0.0592}{n} log(\frac{[M^{n+}_{anode}]}{[M^{n+}_{cathode}]})$$

In our example above, $E = 0.0296V$ for $[Cu^{2+}_{anode}] = 0.1M$ and $[Cu^{2+}_{cathode}] = 1.0M$ (or $Q = 0.1$). Because the potential is low compared to that of a galvanic cell, it does not actually provide any practical value in the context of batteries.

Okay, now we move onto one last type of electrochemical cell called hydrogen fuel cells. **Hydrogen fuel cells** does not use the typical metal elements as the anode and cathode, but rather uses the **hydrogen gas** as the anode and **oxygen gas** as the cathode. It simply utilizes the e^- transfer energy during the process of forming water from these gases.

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$$

When we break this chemical reaction down into components we get the anode reaction (oxidation)

$$H_2(g) \longrightarrow 2\,H^+(aq) + 2\,e^-$$

and the cathode reaction (reduction)

$$\frac{1}{2}O_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2O(l)$$

Here is a schematic of a hydrogen fuel cell.

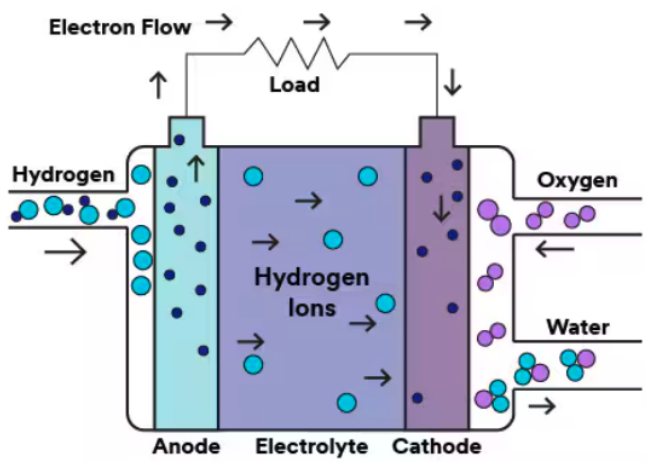

Figure 17: Hydrogen Fuel Cell Diagram

We can quantitatively express the free energy of the hydrogen fuel cell using the same method as we did for the galvanic cell.

$$E^\circ_{cell} = E^\circ_{oxygen} - E^\circ_{hydrogen} = 1.23V - (-0.00V) = 1.23V$$

$$\Delta G^\circ = -(2\,mol)(96485\,C\,mol^{-1})(1.23\,J\,J^{-1}) = -237.35\,kJ$$

Thus the free energy of a hydrogen fuel cell per mole of H_2 is $-237.35\,kJ$.